

Hamza EL BELGHITI

Data & AI Engineer



hamagistral.dev



Hamagistral



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hamza.lbelghiti@gmail.com



Toulouse, France

EXPERIENCE

AIRBUS

September 2025 – August 2026

Data & AI Engineer

Toulouse, France

- Designed an end-to-end ML pipeline using XGBoost to predict customer presentation dates at the A330/A350 FAL, achieving a **MAE of 4.1 days and 87% of predictions within ± 7 days**, improving team planning and resource allocation
- Accelerated the identification of recurring quality non-conformities by developing an LLM-based analysis system for QLBs from final cabin presentations
- Reduced quality teams manual search time by designing a RAG pipeline that classifies QLBs by analyzing their description, automatically retrieving the applicable acceptance criteria (reference, page, conditions) and status (acceptable/non-conforming)
- Maintained and evolved a dashboard centralizing QLB customer quality data for the A330 and A350 programs

Technologies: Skywise (Workbook, Pipeline Builder, Contour, Modeling, Workshop), Python, PySpark, NLP, Gemini, Prompt Engineering

BLENT AI

February 2024 – July 2025

Analytics & AI Engineer

Paris, France

- Consulted and designed an AI MVP for the DGAC to automate the extraction and monthly structuring of data from 300 French aerodromes using an LLM
- Developed an AI chatbot to assist students on the platform and automated the grading of student projects using a RAG system
- Developed Reverse ETL flows for marketing tools such as HubSpot and CustomerIO, enabling the marketing team to target campaigns on customer segments updated daily instead of manually
- Implemented automated ELT pipelines feeding the Customer Data Platform, integrating 5+ sources
- Unified client reporting by building sales and marketing dashboards analyzing customer KPIs for a 5-person team

Technologies: Airbyte, Mage, Airflow, BigQuery, dbt, Metabase, Looker, MongoDB, Hubspot, Open Metadata, Langchain, HuggingFace, Groq, Mistral, Pinecone, PDFplumber, OpenAI API, ChromaDB, Streamlit

3D SMART FACTORY

June 2023 – September 2023

Machine Learning Engineer

Mohammedia, Morocco

- Reduced infrastructure costs by **98% (\$85/month to \$1.3 per 1,000 requests)** by deploying a deep learning model for 3D tooth segmentation via a serverless architecture
- Delivered an interactive 3D visualization interface for exploring tooth segmentations, integrating 3D libraries

Technologies: Vite, React, Tailwindcss, Three, VTK.js, Firebase, FastAPI, Docker, Vercel, AWS

EDUCATION

MASTER MIAE DATA ENGINEERING

University of Toulouse III (France)

Courses: Data Science, Data Engineering, Machine Learning, Big Data, Gen AI, Cloud & DevOps, Agile Projects, Data Governance

2024 - 2026

PROJECT

CRAFTCOURSE : AI-POWERED PERSONALIZED LEARNING PLATFORM | [DEMO](#)

SAAS PROJECT (+500 users)

- Development of a platform that generates visual roadmaps and structured courses personalized to the learner's level, goals, and schedule
- Integration of a conversational AI tutor accompanying the learner in real time with automatic generation of quizzes, summaries, and flashcards
- Implementation of a full gamification system (XP, streaks, leaderboard, verifiable certificates) to maximize learner engagement and retention

Technologies: Next.js, Supabase, pgvector, Gemini API, Hetzner VPS, Webshare, Tailwind, Resend, Vercel, Stripe, Claude Code

SKILLS

Programming: Python, Java, C

Databases: BigQuery, PostgreSQL, Supabase, Redis, MongoDB, Oracle, Vector DBs (pgVector, Pinecone, ChromaDB)

AI & ML: PyTorch, Scikit-learn, LangChain, HuggingFace, RAG, MLflow, AI Agents (CrewAI, n8n, LangGraph), Fine-tuning, MLOps

Web Development: HTML/CSS, React, Next.js, TypeScript, REST APIs, FastAPI, Django, Streamlit

Cloud: Google Cloud, AWS, Skywise

Tools: Linux, Docker, Git, GitHub Actions, Jira, ClickUp

Languages: French, English, Arabic

CERTIFICATIONS

GCP Professional Cloud Architect - [Google](#)

GCP Professional Data Engineer - [Google](#)

Machine Learning - [FreeCodeCamp](#)

Neural Networks and Deep Learning - [DeepLearning.AI](#)

Data Scientist - [DataCamp](#)

Data Engineer - [DataCamp](#)